

Amazon Sales Analysis

1. Project Overview

This project analyzes a large-scale Amazon sales transactional dataset containing **100,000 customer orders** across multiple product categories, brands, payment methods, and geographic locations.

The objective of this analysis is to uncover business insights related to revenue generation, customer purchasing behavior, product performance, cancellations, and regional sales contribution using SQL-based analytical queries.

This project simulates real-world e-commerce business intelligence analysis to support strategic decision-making.

2. Dataset Summary

- Rows: 100,000 - Columns: 20 - Key Features:
- Order details (Order ID, Order Date, Order Status)
- Product details (Product ID, Product Name, Category, Brand, Quantity, Unit Price, Discount)
- Financial metrics (Tax, Shipping Cost, Total Amount)
- Transaction details: (Payment Methods)
- Geographic Information: (City, State, Country)
- Seller Information: (Seller ID)

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary stat

	orderid	orderdate	customerid	customername	productid	productname	category	brand	quantity	unitprice	discount
count	100000	100000	100000	100000	100000	100000	100000	100000	100000.000000	100000.000000	100000.000000
unique	100000	1825	43233	200	50	50	6	10	NaN	NaN	NaN
top	ORD00000001	2022-04-28	CUST023748	Karan Joshi	P00019	LED Desk Lamp	Electronics	ReadMore	NaN	NaN	NaN
freq	1	85	10	556	2098	2098	16853	10204	NaN	NaN	NaN
mean	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	3.001400	302.905748	0.074226
std	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1.413548	171.840797	0.082583
min	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1.000000	5.000000	0.000000
25%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2.000000	154.190000	0.000000
50%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	3.000000	303.070000	0.050000
75%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	4.000000	451.500000	0.100000
max	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	5.000000	599.990000	0.300000

tax	shippingcost	totalamount	paymentmethod	orderstatus	city	state	country	sellerid
100000.000000	100000.000000	100000.000000	100000	100000	100000	100000	100000	100000
NaN	NaN	NaN	6	5	20	13	5	1999
NaN	NaN	NaN	Credit Card	Delivered	Charlotte	TX	United States	SELL01099
NaN	NaN	NaN	35038	74628	5110	24896	70058	78
68.468902	7.406660	918.256479	NaN	NaN	NaN	NaN	NaN	NaN
74.131180	4.324057	724.508332	NaN	NaN	NaN	NaN	NaN	NaN
0.000000	0.000000	4.270000	NaN	NaN	NaN	NaN	NaN	NaN
15.920000	3.680000	340.890000	NaN	NaN	NaN	NaN	NaN	NaN
45.250000	7.300000	714.315000	NaN	NaN	NaN	NaN	NaN	NaN
96.060000	11.150000	1349.765000	NaN	NaN	NaN	NaN	NaN	NaN
538.460000	15.000000	3534.980000	NaN	NaN	NaN	NaN	NaN	NaN

- **Data Quality Validation:** Conducted a comprehensive null value check across the dataset and confirmed that no missing values were found, indicating a complete and analysis-ready transactional dataset.
- **Key Observations:**
 - Electronics category has the highest order frequency
 - Customer names repeat, but customer IDs remain unique identifiers
 - Multiple countries and cities indicate geographically distributed sales
 - Different payment methods create transaction segmentation opportunities
 - Order statuses include delivered and cancelled orders
- **Database Integration:** Connected Python script to MySQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue by Product Category** – Calculated total revenue generated by each category.

	Category	Total_Revenue
►	Books	15261837.01
	Home & Kitchen	15163939.36
	Clothing	15253397.5
	Toys & Games	15216684.99
	Sports & Outdoors	15345571.88
	Electronics	15584217.18

2. **Top Revenue-Generating Products** – Identified products contributing the highest total revenue.

	Products	Total_Revenue
►	Memory Card 128GB	1935138.4
	LED Desk Lamp	1921948.41
	Mechanical Keyboard	1906963.54
	Electric Kettle	1905751.79
	Smartwatch	1901275.59
	Dress Shirt	1896765.36
	Water Bottle	1895467.95
	Gaming Mouse	1895103.98
	Kids Toy Car	1891082.43
	Jeans	1882591.82

3. **Top Customers by Total Spending** – Found products with the highest average review ratings.

	Customer_Name	Total_Spending
►	Pooja Patel	5436.04
	Neha Sharma	5124.43
	Vikas Mehta	5003.65
	Arjun Kapoor	4773.18
	Pooja Joshi	4755.48
	Sneha Reddy	4730.47
	Priya Kapoor	4266.02
	Vivaan Kapoor	4253.27
	Sahil Patel	4239.18
	Ariun Kumar	4222.62

4. **Monthly Revenue Trend Analysis** – Analyzed revenue trends over time grouped by month and year.

	Month	Total_Revenue
►	Jan-2020	1637069.4
	Feb-2020	1339388.21
	Mar-2020	1587125.66
	Apr-2020	1588347.29
	May-2020	1533834.17
	Jun-2020	1500269.29
	Jul-2020	1552981.94
	Aug-2020	1640874.93
	Sep-2020	1554682.21
	Oct-2020	1515403.62

5. **Highest Revenue Payment Method** – Compared payment methods by total revenue contribution.

	Payment_Method	Total_Revenue
▶	Credit Card	32122158.69

6. **Revenue Contribution by City** – Analyzed city-wise revenue performance.

	City	Total_Revenue
▶	Charlotte	4747701.73
	Dallas	4730108.67
	San Jose	4669614.33
	Seattle	4660962.17
	Philadelphia	4650673.88
	Denver	4638498.97
	Austin	4638403.18
	Chicago	4632847.44
	Los Angeles	4620629.78
	Columbus	4615442.32

7. **Cancellation Rate Analysis**– Calculated cancellation percentage overall and by category.

	Total_Orders	Cancelled_Orders	Cancellation_Rate
▶	100000	3028	3.03

	category	Total_Orders	Cancelled_Orders	Cancellation_Rate
▶	Books	16752	500	2.98
	Home & Kitchen	16610	508	3.06
	Clothing	16439	467	2.84
	Toys & Games	16542	512	3.10
	Sports & Outdoors	16804	497	2.96
	Electronics	16853	544	3.23

8. **Brand Revenue Performance** – Compared brands by total revenue generated.

	brand	Total_Revenue
►	CoreTech	9343720.59
	KiddoFun	9324749.55
	ReadMore	9278406.63
	UrbanStyle	9249027.45
	Zenith	9239960.39
	Apex	9147604.72
	NexPro	9078824.97
	FitLife	9061444.68
	BrightLux	9056816.11
	HomeEase	9045092.83

9. **Average Order Value by Category** – Measured average spending across categories.

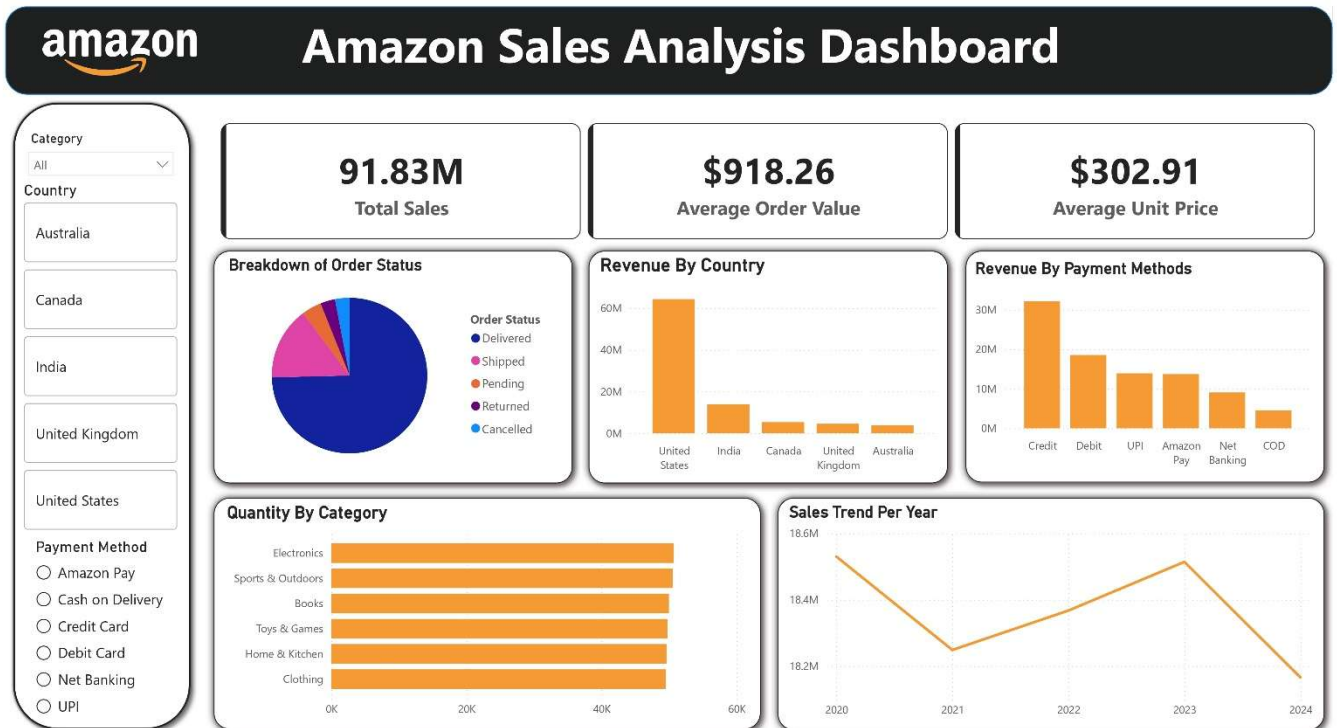
	Category	Avg_Order_Value
►	Clothing	927.88
	Electronics	924.71
	Toys & Games	919.88
	Sports & Outdoors	913.21
	Home & Kitchen	912.94
	Books	911.05

10. **Product Demand Analysis**– Ranked products by total quantity sold.

	Product	Quantity
►	LED Desk Lamp	6344
	Water Bottle	6275
	Memory Card 128GB	6240
	Router	6202
	Board Game	6200
	Microphone	6196
	Gaming Mouse	6170
	Electric Kettle	6165
	Mechanical Keyboard	6161
	Vacuum Cleaner	6139

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



6. Business Recommendations

- **Optimize Payment Channels** – Since credit card transactions generate the highest revenue, optimize promotions around preferred payment channels.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Improve Cancellation Management** – Investigate reasons behind category-specific cancellations to reduce revenue leakage.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Brand Partnership Strategy** – Collaborate more aggressively with top-performing brands.